

HORIZON-WIDERA-2023-ERA-01-10: Support to the implementation of an EU Manifesto for STE(A)M education and research and innovation career paths to tackle gender inequalities in the ERA

Specific conditions	
<i>Expected EU contribution per project</i>	The Commission estimates that an EU contribution of around EUR 2.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
<i>Indicative budget</i>	The total indicative budget for the topic is EUR 2.00 million.
<i>Type of Action</i>	Coordination and Support Actions

Expected Outcome: Projects are expected to contribute to the following expected outcomes:

- Common principles and joint commitment on gender-inclusive STEAM education, research and innovation among a wide range of R&I actors in Member States and Associated Countries;
- Network of STEM-oriented businesses, secondary and higher education institutions, research organisations, informal science education establishments, and civil society organisations, supporting knowledge sharing, mutual learning and structural change towards gender equality and diversity at their organisations;
- Increased cooperation between relevant R&I actors from academia, the private sector, and national administrations to foster women and girls’ participation in STEM studies and careers through a STEAM approach.

Scope: Persisting gender gaps, particularly in STEM fields, are hindering the potential of the European Research Area (ERA) to meet pressing challenges like the green and digital transitions. According to She Figures 2021, women outnumber men at Bachelor and Master levels overall, but are still underrepresented as doctoral graduates in STEM fields, including physical sciences (38%), ICT (20.8%), engineering (27%), and mathematics (32.5%).⁷³ In order to increase the uptake in scientific careers, feed the talent pipeline, and counteract women’s underrepresentation, a STEAM approach, featuring creative thinking, applied arts (the “A” in STEAM), and the teaching of science in political, environmental, socio-economic

⁷³ European Commission, Directorate-General for Research and Innovation, She Figures 2021: gender in research and innovation: statistics and indicators, Publications Office, 2021, <https://data.europa.eu/doi/10.2777/06090>

and cultural contexts, can prove particularly useful. By highlighting the social impact and market relevance of research and innovation, the STEAM approach would also strengthen the appeal of scientific careers.

In line with the European Strategy for Universities and the Digital Education Action Plan 2021-2027⁷⁴, this action aims to support the cooperation between STEM-oriented businesses, secondary and higher education institutions, research organisations, as well as informal science education establishments, NGOs, and civil society organisations, committed to fostering gender-inclusive career paths in the STEM fields through a STEAM⁷⁵ approach.

More specifically, the action should ensure the implementation of a European Manifesto for STEAM education and career paths to tackle gender inequalities among relevant R&I actors, through the adoption of common principles and actions on gender-inclusive STEAM education, research and innovation. Particular attention should also be paid to addressing bottleneck moments and structural barriers in young researchers' careers. The implementation of this manifesto should build on the STE(A)M roadmap(s) for science education, developed by projects under the call topic HORIZON-WIDERA-2021-ERA-01-70, and seek synergies with the ERA Talent Platform initiative and the Digital Education Hubs.

To support relevant R&I actors, including umbrella organisations, with the implementation of an EU Manifesto for STEAM education to tackle gender inequalities, proposals are expected to cover the following:

- Development of a network of national R&I and education actors from Member States and Associated Countries, involved in STEAM education, research and innovation, to support best practice sharing, mutual learning, and transformative actions to tackle gender inequalities at education institutions and R&I organisations;
- Setting-up of principles and a joint commitment on gender equality and diversity in STEM education, research and innovation at organisational level, in line with the ERA, EEA and the European Innovation Agenda;
- Communication and awareness-raising on the deconstruction of gender stereotypes and unconscious biases in science and the benefit of STEAM education, research and innovation;
- Support STEM skills-development and learning opportunities for all genders through the use of arts, social sciences and humanities, including e.g., mentoring sessions, non-cost summer schools, workshops, and traineeship opportunities, and involving inter alia start-

⁷⁴ Commission Communication on a European Strategy for Universities, 4.2 Foster diversity, inclusiveness and gender equality, <https://education.ec.europa.eu/document/commission-communication-on-a-european-strategy-for-universities>; Digital Education Action Plan 2021 -2027, Action 13: Women participation in STEM, <https://education.ec.europa.eu/focus-topics/digital/education-action-plan>.

⁷⁵ Science, technology, engineering, (the arts) and mathematics (STEAM) is a multidisciplinary set of approaches to education removing traditional barriers between subjects and disciplines to connect STEM and ICT education with the arts, humanities, and social sciences.